Amazon Brazil: E-Commerce SQL Analysis

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Project Overview

This report answers fourteen business questions about orders, sellers, customers, payments, reviews and delivery on a Brazilian online marketplace. Each question is framed as a request from a business team, such as logistics, finance, CRM or merchandising. For every question the report gives the business problem, the question itself, the SQL query used, a short summary of the results and the business insight that follows from them.

Dataset

The data is the public Amazon Brazil e-commerce dataset, loaded into MySQL. It has 11 tables: 99,441 orders, 112,650 order items, 103,886 payment records, 99,224 reviews, 3,095 sellers, 32,951 products and 71 category translations, plus customer, geolocation and seller-lead tables. Orders run from September 2016 to September 2018 and all money values are in Brazilian reais (R$).

Approach

All queries were written in MySQL and run in MySQL Workbench. The techniques used include inner and left joins, anti-joins for "never happened" questions, grouping with DISTINCT counts to avoid double counting, common table expressions, and window functions for running totals and the median.

Questions Covered

Q	Topic	Table(s) used
1	Seller-State Order Volume	orders, order_items, sellers
2	Cumulative Revenue by Product Category	order_items, orders, products
3	Payment Methods: Usage and Average Value	order_payments
4	Highest-Spending Customer	customers, orders, order_payments
5	Average Review Score by Category	order_reviews, order_items, products
6	Orders per Customer by State	customers, orders
7	Sellers Who Never Fulfilled an Order	leads_closed, order_items, orders
8	Top 5 Categories by Revenue	order_items, products, translation
9	Median Delivery Time	orders
10	Products That Were Never Ordered	products, order_items
11	Sellers Above the Average Order Count	sellers, order_items, orders
12	Review Score by Customer State	order_reviews, orders, customers
13	Customers Who Never Left a Review	customers, orders, order_reviews
14	Peak Order Month	orders

Q1. Seller-State Order Volume

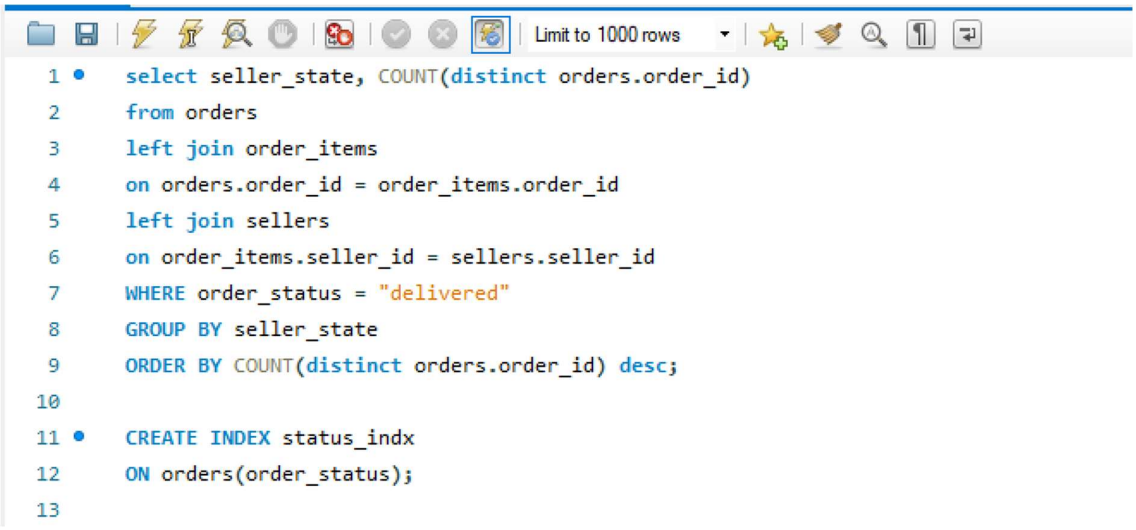
Business Problem

The logistics team is planning its next round of regional fulfilment-centre investments. Before committing budget, they need to know which states generate the most order volume on the seller side, so that infrastructure goes where the demand is.

Question

Find the total number of orders fulfilled by each seller state.

SQL Query



Result Summary

Only delivered orders were counted, and each order was counted once per seller state. São Paulo (SP) leads by a wide margin with 68,641 orders, followed by Minas Gerais (MG) with 7,735, Paraná (PR) with 7,512, Rio de Janeiro (RJ) with 4,227 and Santa Catarina (SC) with 3,603. The 22 states returned add up to 96,994 state-level order counts; this is slightly above the number of delivered orders because an order with sellers in two states appears in both.

SP alone accounts for about 71% of the total, and the top three states (SP, MG, PR) for about 86%. At the bottom, Amazonas (AM) has 3 orders and Pará (PA) has 8.

Business Insight

Seller-side fulfilment is heavily concentrated in the South-East and South. Any new fulfilment-centre investment has the strongest volume case in SP, with MG, PR, RJ and SC as the next tier. The very low counts in northern and north-eastern states show that the seller base there is thin, so opening facilities in those regions would also require seller acquisition, not just infrastructure. The heavy dependence on SP is also a concentration risk worth monitoring.

Q2. Cumulative Revenue by Product Category

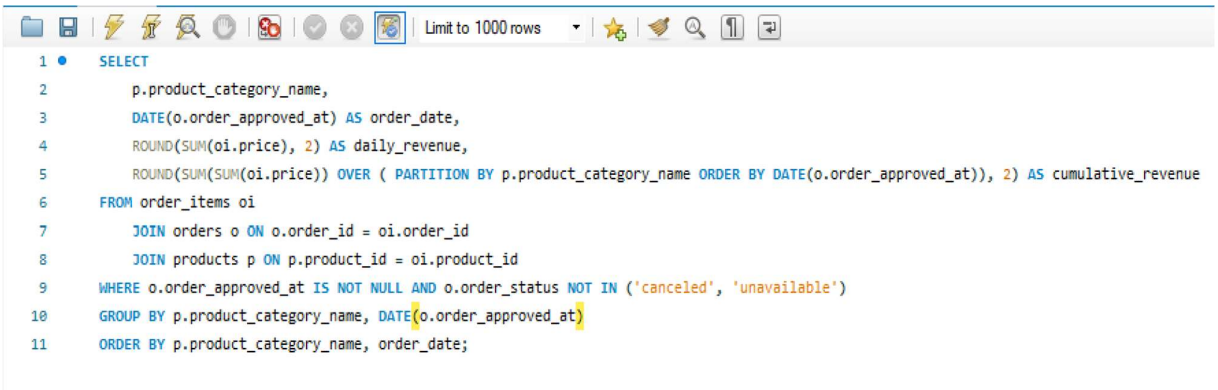
Business Problem

The growth team wants to see how revenue builds up within each category across the order timeline. The running total helps them spot which categories reach revenue targets early and which pick up late, which is useful for planning flash sales and push notifications.

Question

For each product category, calculate the cumulative revenue generated as orders come in over time.

SQL Query



Result Summary

The result is a daily running total per category, covering 73 named categories plus an uncategorised group, from September 2016 to September 2018 (18,645 rows). Revenue is the item price (freight excluded), based on payment approval date, with cancelled and unavailable orders removed.

By the last date in the data, health and beauty (beleza_saude) is ahead at about R$1.26 million, followed by watches and gifts (relogios_presentes) at R$1.20 million, bed, bath and table (cama_mesa_banho) at R$1.04 million, sports and leisure (esporte_lazer) at R$0.98 million and computers and accessories (informatica_acessorios) at R$0.90 million.

Business Insight

The running totals show which categories carry the platform's revenue and how quickly each one has grown, which the growth team can use to time flash sales around categories that are still climbing. Because the total runs across the full two-year timeline rather than restarting each month, it does not yet answer the early-month versus late-month question in the brief. Restarting the running total every month would answer it, and the output is best viewed as a line chart per category in a BI tool.

Q3. Payment Methods: Usage and Average Value

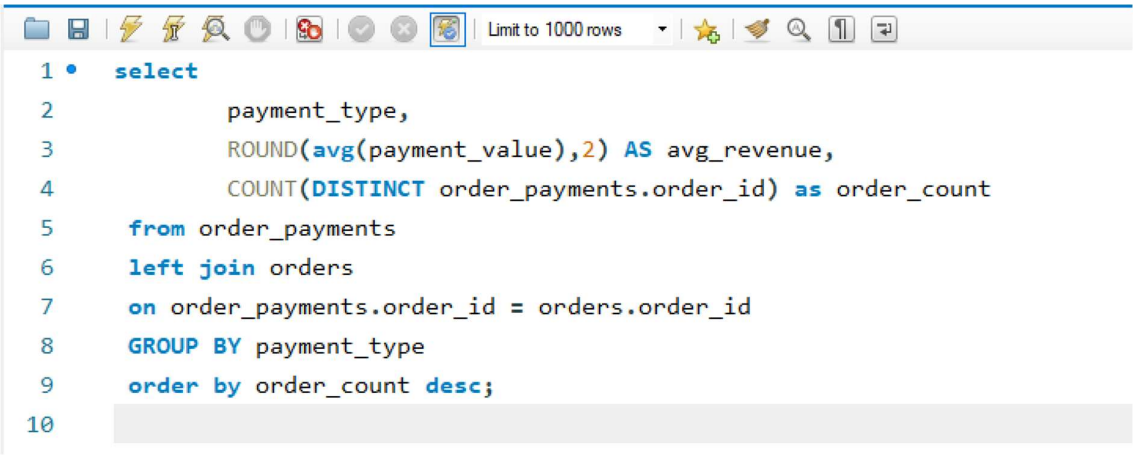
Business Problem

The finance and partnerships team is renegotiating contracts with payment providers. They need to know which payment method dominates the platform and whether high-value orders cluster around a specific payment type.

Question

Which payment method do customers use the most, and what is the average order value for each payment type?

SQL Query



Result Summary

Credit card is the most used method with 76,505 orders and the highest average payment of R$163.32. Boleto follows with 19,784 orders (R$145.03), then voucher with 3,866 orders (R$65.70) and debit card with 1,528 orders (R$142.57). Three orders have an undefined payment type with a value of zero.

Credit card covers roughly 75% of orders when counted by payment method (an order paid with two methods is counted under each). Averages are per payment record, not per whole order.

Business Insight

Credit card dominates both in volume and in value, so it should be the focus of the provider negotiations, with boleto as the clear second priority. High-value payments do lean towards credit card, which is about 13% higher on average than boleto. The low voucher average is probably because vouchers often cover only part of an order. The three undefined records are a small data-quality issue to flag.

Q4. Highest-Spending Customer

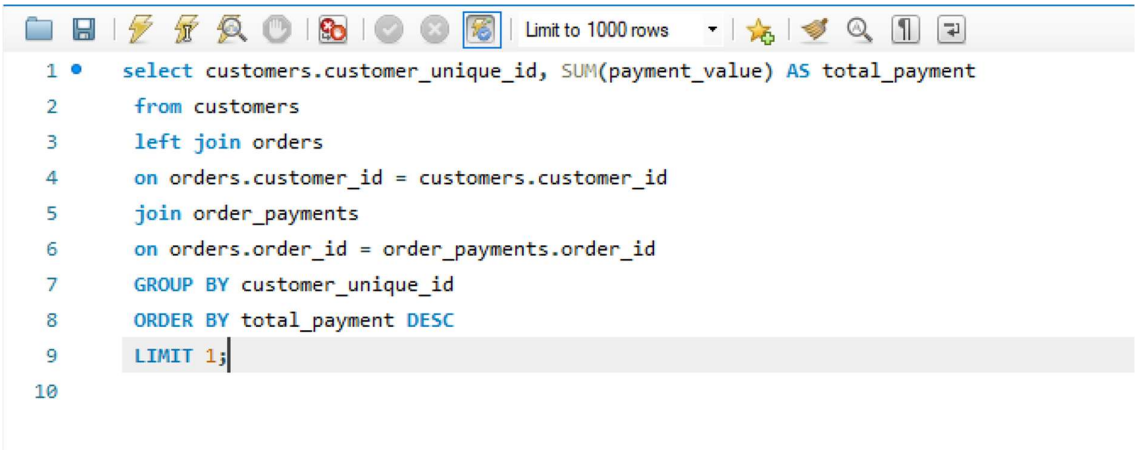
Business Problem

The CRM team is launching the first premium loyalty tier. The first step is identifying the single highest-spending customer, both to onboard them personally and to understand what a top-tier customer profile looks like.

Question

Find the customer who has spent the most money across all their orders.

SQL Query



Result Summary

The top customer (customer_unique_id 0a0a92112bd4c708ca5fde585afaa872) spent a total of R$13,664.08. Spending was summed across all of the customer's orders using the unique customer ID, so repeat orders from the same person are combined.

For scale, this is about 84 times the average credit-card payment of R$163.32 seen in Q3.

Business Insight

One customer stands far above a typical payment, which makes them an obvious first invitation to the premium tier. Before onboarding, the CRM team should check whether this spend comes from many orders (true loyalty) or a few very large purchases, and which categories and state it comes from, to build the top-tier profile described in the brief.

Q5. Average Review Score by Category

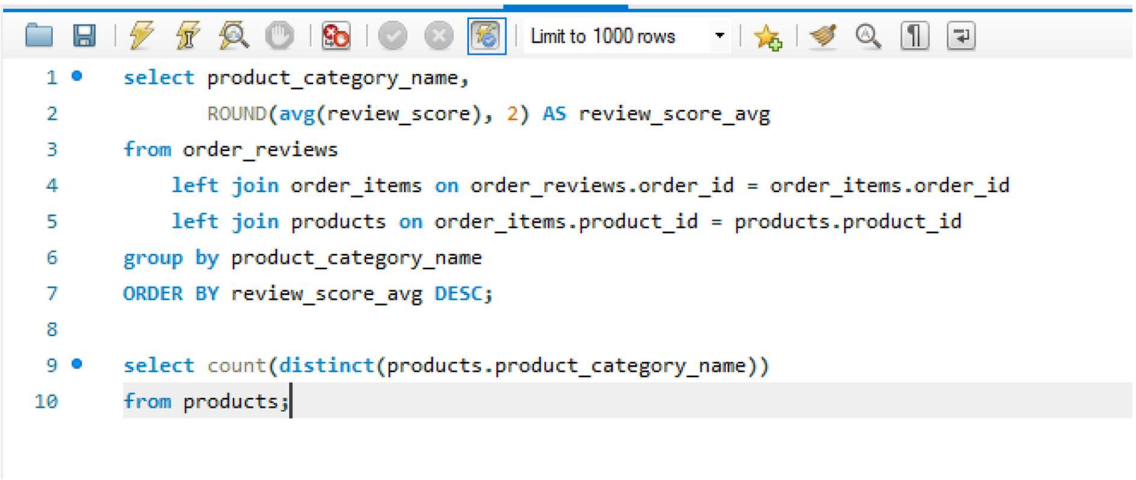
Business Problem

The customer experience team tracks satisfaction at category level. A low average score often signals a fulfilment problem, a quality problem or a mismatch between the listing and the product, and this analysis helps them decide which categories need intervention first.

Question

Find the average review score for each product category.

SQL Query



Result Summary

The output has 74 rows: 73 named categories and one uncategorised group (average 3.16). The best-rated categories are cds_dvds_musicais (4.64), fashion_roupa_infanto_juvenil (4.50) and livros_interesse_geral (4.45). The lowest are seguros_e_servicos (2.50), fraldas_higiene (3.26), portateis_cozinha_e_preparadores_de_alimentos (3.27), pc_gamer (3.33) and moveis_escritorio (3.49).

Business Insight

The spread of more than two points between the best and worst category shows that satisfaction varies a lot by product type. The low-scoring categories are the first candidates for review, starting with a check of delivery times and product descriptions. Small categories can swing an average easily, so the number of reviews behind each score should be checked before acting. Fixing missing category data would also remove the uncategorised group, which scores below most named categories.

Q6. Orders per Customer by State

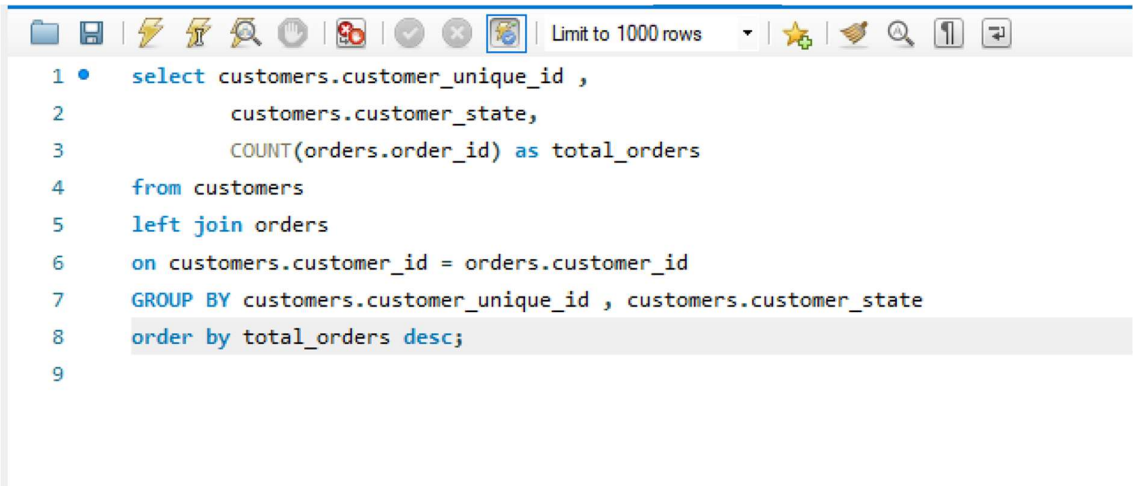
Business Problem

The regional marketing team is planning state-level campaigns. They want to know how order frequency varies by customer location, so they can target low-frequency states with acquisition offers and reward high-frequency states with loyalty perks.

Question

Find the total number of orders placed by each customer, broken down by the state they live in.

SQL Query



Result Summary

Each row is one customer (unique ID) with their state and number of orders. The exported results show the top 1,000 customers, all with two or more orders, so this is a partial view of the full output. The most active customer is in SP with 17 orders, followed by another SP customer with 9, and then customers with 7 or fewer.

Within the exported rows, SP accounts for 420 customers, RJ for 144, MG for 120, RS for 60 and PR for 54.

Business Insight

Order frequency falls away sharply after the first two customers, which suggests that repeat buying is limited and that loyalty perks would reach a small group. For the state-level campaigns, the customer-level output should be summarised by state (customers per state and average orders per customer) to rank high- and low-frequency states properly.

Q7. Sellers Who Never Fulfilled an Order

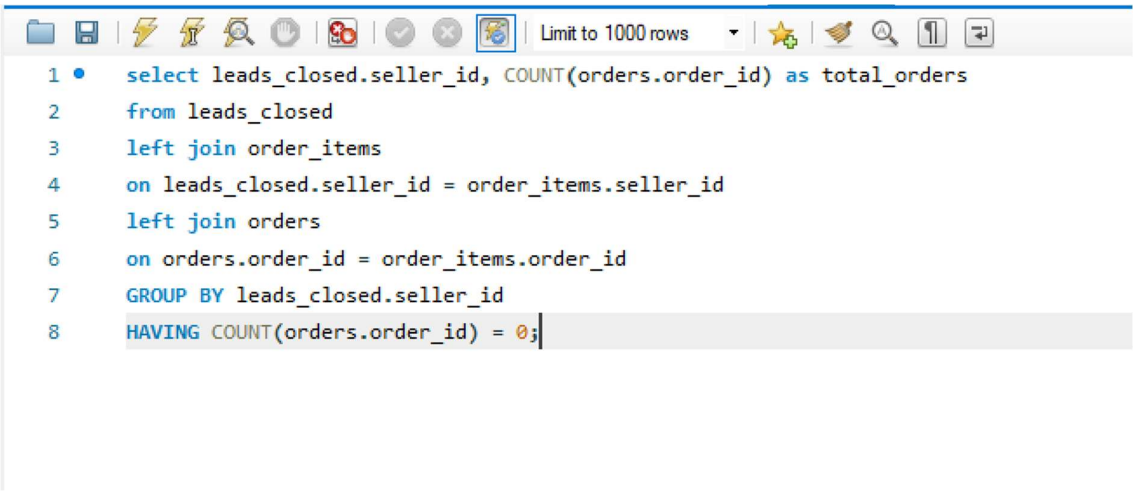
Business Problem

The seller growth team runs quarterly re-engagement campaigns for dormant sellers. To build the list, they need every seller account that exists in the system but has zero fulfilled orders, since these accounts need a nudge or a review for removal.

Question

Identify sellers who registered on the platform but have never fulfilled a single order.

SQL Query



Result Summary

462 sellers from the converted-leads table have no orders at all. Out of the 842 converted leads, this is about 55%, meaning more than half of the sellers who were signed up never made a sale.

Business Insight

This points to a gap between seller sign-up and first sale. The 462 accounts are a ready-made list for the re-engagement campaign. The leads table also records business segment, lead type and business type, so comparing these dormant sellers with active ones could show which kinds of leads fail to start selling and improve how new sellers are onboarded.

Q8. Top 5 Categories by Revenue

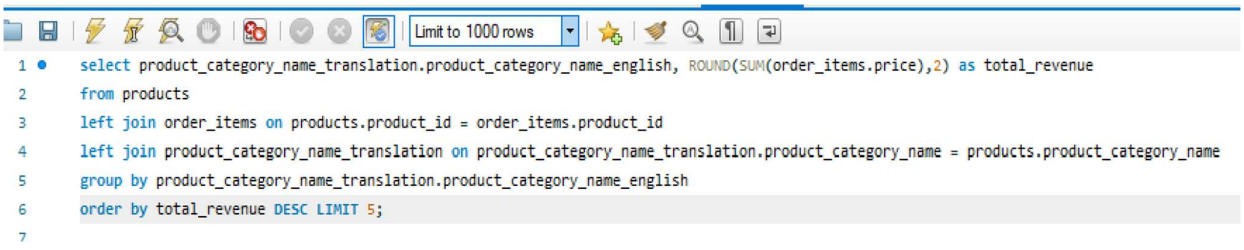
Business Problem

Before every major sales event, such as Black Friday, Dia das Mães and Cyber Monday, the merchandising team decides which categories get homepage space and promotional budget. Total revenue by category is the main input for that decision.

Question

Find the top 5 product categories by total revenue.

SQL Query



Result Summary

The top five categories are health_beauty (R$1,258,681.34), watches_gifts (R$1,205,005.68), bed_bath_table (R$1,036,988.68), sports_leisure (R$988,048.97) and computers_accessories (R$911,954.32). Together they bring in about R$5.40 million. Revenue is the sum of item prices, so freight is excluded.

Business Insight

The top two categories are close, with health_beauty only about 4.5% ahead of watches_gifts, so both deserve prime homepage placement. The five categories cover different shopper needs (personal care, gifting, home, sport and technology), which allows a varied promotional line-up. Gift-oriented categories are a natural fit for occasion-based events.

Q9. Median Delivery Time

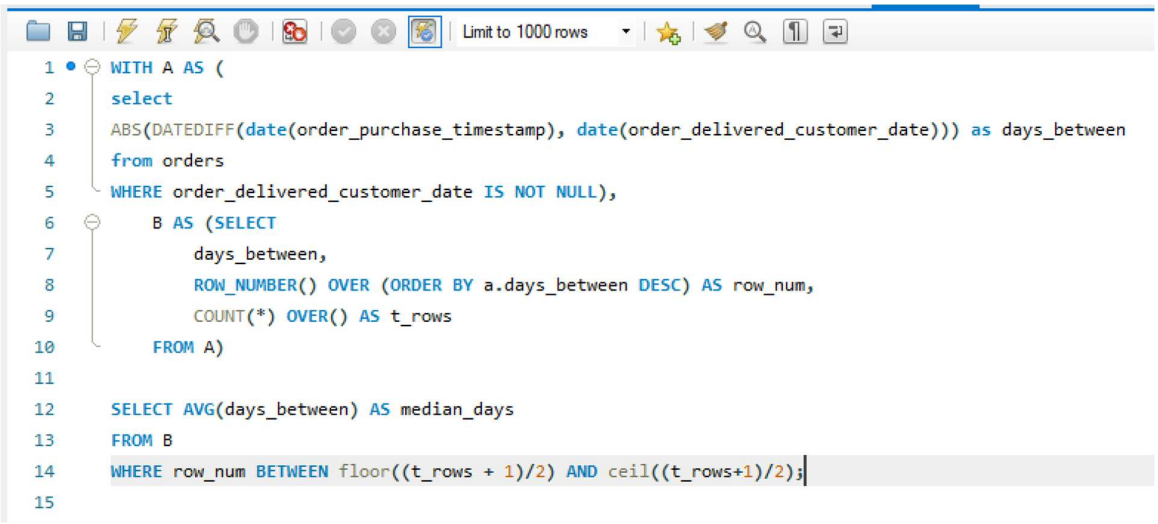
Business Problem

The logistics director wants to know how long customers typically wait. The median is preferred over the average because a few extremely delayed orders to remote parts of Brazil's interior would pull the average up and make performance look worse than it is for most customers.

Question

Find the median delivery time (in days) between order placement and actual delivery.

SQL Query



Result Summary

The median delivery time is 10 days, measured from order purchase to actual delivery for orders that have a delivery date. This means half of delivered orders arrived within 10 days.

Business Insight

Ten days is the realistic baseline for what a typical customer experiences. To manage the long tail that the median hides, the logistics team should also track a high percentile (for example the 90th) and break delivery time down by state, which would link directly to the regional satisfaction results in Q12.

Q10. Products That Were Never Ordered

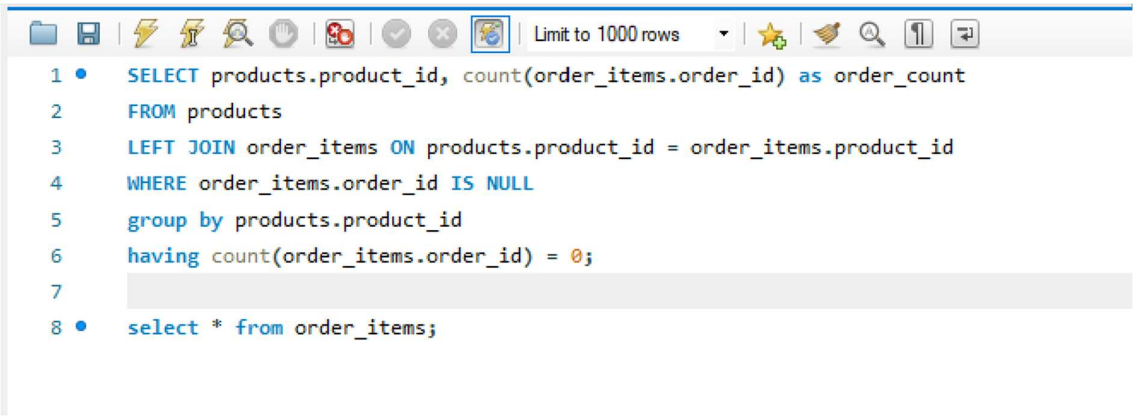
Business Problem

The catalogue operations team does a quarterly clean-up of dead listings. Products that never receive an order waste search index space, confuse customers and inflate category pages, so the team needs an exact list to review, reprice or remove.

Question

Find all products that have never been ordered.

SQL Query



Result Summary

The query returned no rows, so every product in the products table appears in at least one order.

Business Insight

In this dataset there are no dead listings to remove. The products table appears to contain only products that have sold. On a live catalogue, the clean-up would need the full list of active listings, or a looser definition of dead, such as no orders in the last 6 or 12 months or no successfully delivered orders.

Q11. Sellers Above the Average Order Count

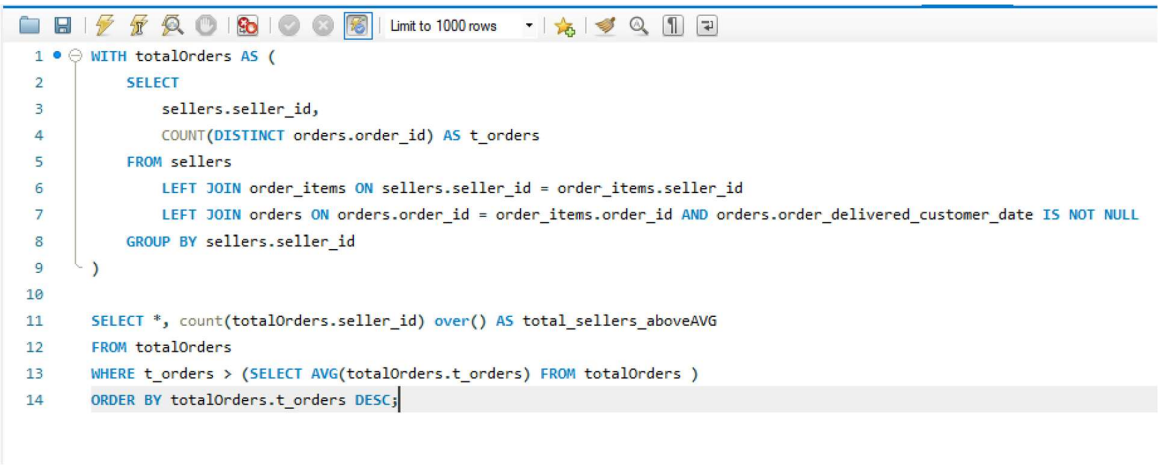
Business Problem

The seller success team is building an invite list for the "Top Seller" badge. Only sellers who perform above the platform average qualify, and the team also wants a numerical baseline for what above-average performance means.

Question

Find sellers who have fulfilled more orders than the average seller on the platform.

SQL Query



Result Summary

593 of the 3,095 sellers (about 19%) are above the average, counting delivered orders only and including sellers with zero orders in the average. The busiest seller handled 1,819 delivered orders and the lowest qualifying seller handled 32. The median qualifying seller handled 73 orders, and the top 10 sellers together handled 13,678.

Business Insight

The badge would go to roughly one seller in five. Because a handful of very large sellers pull the average up, the range among qualifiers is huge (32 to 1,819 orders), and a single badge treats them all the same. Tiered badges (for example silver, gold and platinum) would recognise the biggest sellers separately and give mid-sized sellers a realistic target.

Q12. Review Score by Customer State

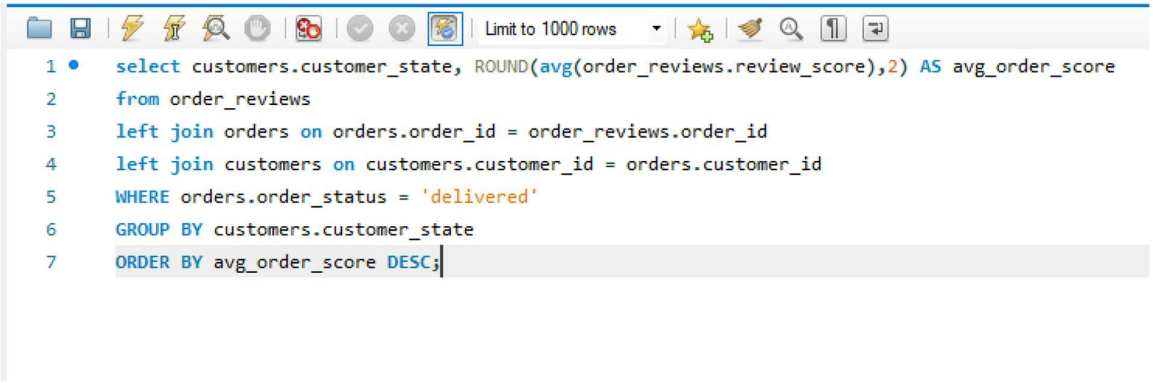
Business Problem

Last-mile delivery varies significantly across Brazil. Deliveries to São Paulo and Rio are generally faster and smoother than those to the North and North-East. This analysis helps the CX team link delivery destination with satisfaction so that logistics partners can be pushed to improve in weaker states.

Question

Find which Brazilian states have the highest average customer review score for orders delivered there.

SQL Query



Result Summary

Across 27 states, SP has the highest average score (4.25), followed by PR (4.24), AP (4.24), AM (4.22), MG (4.19) and RS (4.19). The lowest are MA and AL (both 3.84), RR (3.90), and SE and PA (both 3.91). Other low scores include RJ (3.96), CE (3.94) and BA (3.93).

Business Insight

The pattern broadly supports the brief: scores are strongest in the South-East and South and weakest in the North and North-East, with a gap of about 0.4 points between the best and worst state. Two exceptions, AP and AM, rank high but have low order volumes, so their averages may be less reliable. RJ is the more important finding, because it is a high-volume state scoring below 4.0, which makes it a priority for logistics partners.

Q13. Customers Who Never Left a Review

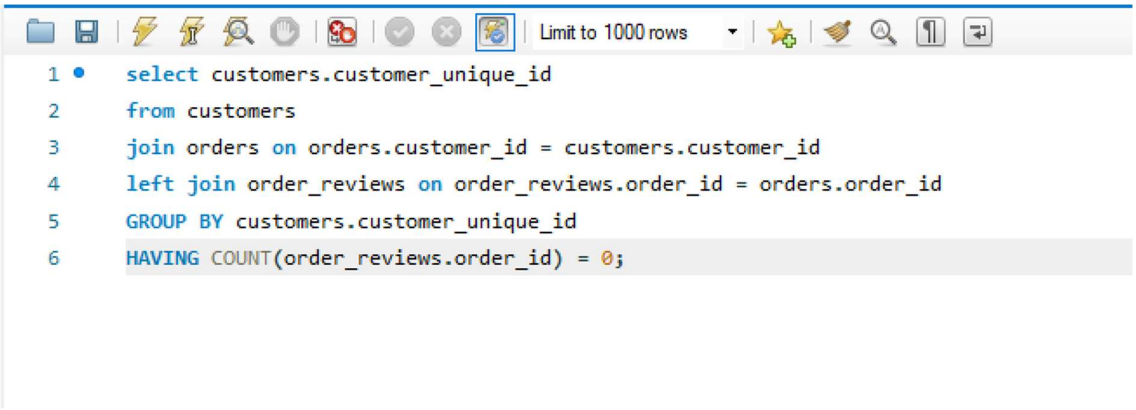
Business Problem

The post-purchase engagement team wants more reviews on the platform, because more reviews improve product discoverability and buyer trust. The first step is identifying customers who buy but never review, so they can be targeted with reminder emails and in-app nudges.

Question

Identify customers who have placed orders but never left a review.

SQL Query



Result Summary

716 customers placed at least one order and none of their orders has a review, which is under 1% of the roughly 99,000 customer records.

Business Insight

The review gap is small, so a reminder campaign aimed at non-reviewers would reach a small group. The bigger opportunity is the content of the reviews: according to the data dictionary, about 59% of reviews have no written comment. Prompting reviewers to add a short comment would do more for buyer trust than chasing the 716 customers who left no rating.

Q14. Peak Order Month

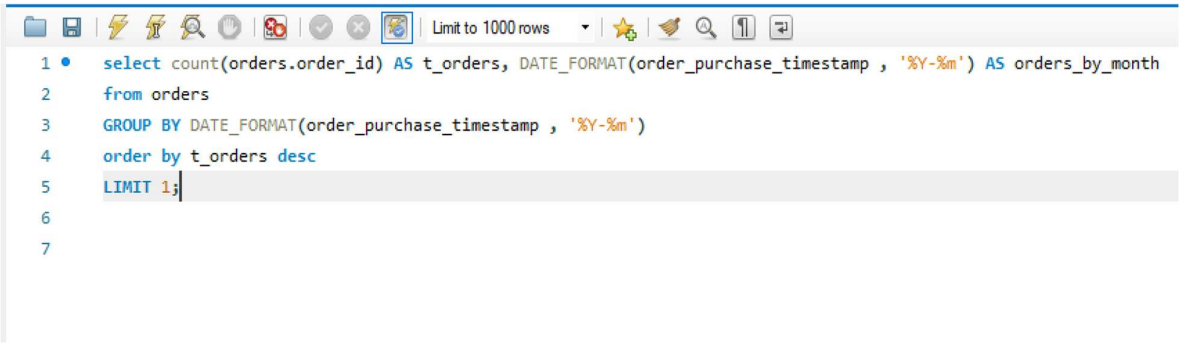
Business Problem

The demand forecasting team needs to know the platform's historical peak order month. This drives how much warehouse capacity to reserve, how many temporary delivery staff to hire and when to start marketing for the next peak season.

Question

Find the month with the highest number of orders placed across the entire platform.

SQL Query



Result Summary

The peak month is November 2017 with 7,544 orders placed. Orders are counted by purchase date, grouped by year and month.

Business Insight

The peak coincides with Black Friday (24 November 2017), so the main driver is most likely the sales event. Capacity planning, extra delivery staff and marketing should be in place before November. Looking at the months either side of the peak would show how long the surge lasts and how quickly demand falls back, which helps size temporary capacity.

Conclusion

The fourteen queries give a consistent picture of the marketplace. The main findings are grouped below.

Key Findings

- **Concentration.** São Paulo accounts for about 71% of seller-side orders, credit card for about 75% of orders by payment method, and five categories (led by health_beauty) bring in about R$5.4 million.
- **Seller health.** 462 of 842 converted leads (55%) never sold anything, and only 593 of 3,095 sellers (19%) are above the average order count, with the busiest seller handling 1,819 orders.
- **Customer experience.** The median delivery time is 10 days. Review scores range from 3.84 (MA, AL) to 4.25 (SP) across states, and from 2.50 to 4.64 across categories.
- **Demand and customers.** Orders peaked in November 2017 at 7,544, around Black Friday. Repeat buying is limited (the most active customer placed 17 orders) and only 716 customers never left a review.